

Process Characterization Report

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCR-005

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; Cpk, one-sided process capability index; CQA, critical quality attribute; CV, column volume; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; ICH, International Council for Harmonisation; KPP, key process parameter; LoQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; ppm, parts per million; qPCR, quantitative polymerase chain reaction; RSD, relative standard deviation; RSM, response surface methodology; SD, standard deviation; SDM, scale-down model; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; WC-CPP, well-controlled critical process parameter.

Executive summary

Protein A chromatography is the capture step of the A-Mab drug substance process and the first chromatographic operation in the purification train. The step binds A-Mab from the clarified harvest on an immobilized Protein A resin, removes host cell protein (HCP) and DNA in the flow-through and the wash, and elutes the antibody at low pH into the low-pH viral inactivation step. Characterization was executed under the paired plan PCP-005 on a scale-down model qualified against manufacturing-scale performance (SOP-1001). The work was executed to the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), to the enhanced development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012) and to the risk basis recorded in RA-001. Commercial manufacture of the drug substance runs at 15,000 L (CMC Biotech Working Group 2009).

In total 6 process parameters were assessed at this step. Of these, 4 were studied in a two-level factorial screening design of 19 runs and then in a face-centred central composite design of 28 runs, while the remaining 2 (operating temperature and bed height) were assessed one parameter at a time. On the evidence set out in §9, 2 parameters were classified as well-controlled critical process parameters, 2 as key process parameters and 2 as general process parameters. No parameter at this step was classified as a critical process parameter.

The step sets one critical quality attribute, leached Protein A, and is the principal clearance step for host cell protein and residual DNA (§2.2). Pool host cell protein is the response that bounds the operating region. Because the response-surface model for that response is adequate and predictive ($R^2 = 0.989$, adjusted $R^2 = 0.976$, predicted $R^2 = 0.932$), the design space in §6 is derived from it. Over the characterized region, the model places 26.6% of the evaluated settings above the in-process limit of 10,724 ng/mg. The same limit was exceeded by 9 of the 47 pools assayed across the two designs, every one of them at the low edge of the elution pH range. The settings that produced those pools mark an edge of failure for the step.

The normal operating range is not wholly inside the acceptable region. That range includes a corner at protein load 40 g/L resin, elution buffer pH 3.4, load flow rate 150 cm/hr and end of pool collect 2.3 CV, where the model predicts 13,553 ng/mg against a limit of 10,724 ng/mg. However, the in-process limit itself carries an assurance factor of 4.5 below the pool concentration above which the downstream train can no longer bring a batch to specification. Even at that corner the drug substance would carry 28 ng/mg against a specification of 100 ng/mg. The operating region is therefore stated in §6 as a joint constraint on protein load and elution buffer pH.

Leached Protein A produced no model of predictive value. Neither the screening fit nor the response-surface model reaches significance ($p = 0.87$ and $p = 0.36$), with a negative predicted coefficient of determination in both cases. For this reason the

claim for this attribute is made from the assayed pools without a model. All 47 pools met the acceptance criterion of 5 ppm, and the highest of them reached 4.61 ppm.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches operated within the normal operating range, and all three attributes this step governs meet the drug-substance criteria (§8). The tightest of the three attributes is Host Cell Protein (HCP), at Cpk 6.1 against a limit of 100 ng/mg, with a simulated mean of 15.2 ng/mg. Two deviations were recorded during execution and are reported in §13. Both were investigated, dispositioned as retained and assessed as having no effect on the fitted models, on the operating region or on the classifications.

The step therefore delivers a bounded operating region for the two parameters that govern pool host cell protein, a leached Protein A level inside the criterion at every setting studied and the capture-stage share of host cell protein and DNA clearance. Protein A chromatography does not by itself bring host cell protein to the drug-substance specification. The remaining clearance is delivered by cation exchange (PCR-007) and anion exchange (PCR-008), while no viral clearance is claimed for this step. These results are rolled up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody that is manufactured by fed-batch mammalian cell culture and purified on an established platform train (CMC Biotech Working Group 2009). The train runs Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration with diafiltration. Protein A chromatography is Step 5 of that train and the first purification step after harvest and clarification (PCR-004).

The step is operated in bind and elute mode on an immobilized Protein A affinity resin. Clarified harvest is loaded at 30 g/L resin onto a bed of 20 cm at a linear velocity of 200 cm/hr. The antibody binds through the Fc region while host cell protein, DNA and media components pass into the flow-through and are further reduced by the post-load wash. Bound antibody is released by a low-pH elution buffer at pH 3.55, with the eluate collected to 2.6 column volumes on the descending edge of the peak. Because the low-pH eluate is also the feed to the viral inactivation step, the elution condition serves two purposes at once.

Three impurity attributes are governed here. Leached Protein A is formed at this step and at no other, since it is the affinity ligand released from the resin under the elution condition. Host cell protein and residual DNA are formed in the production bioreactor (PCR-003) and are cleared here for the first time. Product quality attributes formed upstream, such as the glycan and charge variants, pass through the step unaltered, because affinity capture does not discriminate between them (CMC Biotech Working Group 2009).

1.2 Objectives

This report records the executed characterization of Step 5 and has five objectives:

- (i) quantify the relationship between each studied parameter and the responses that matter at this step, namely pool host cell protein, step yield and leached Protein A;
- (ii) define the multivariate operating region over which the governed attributes remain within their criteria;
- (iii) establish proven acceptable ranges for each parameter against each governed attribute;

- (iv) classify every parameter of the step against the decision logic in SOP-4001; and
- (v) justify the Stage 2 ranges and the contribution this step makes to the control strategy.

1.3 Related documents

The documents that scope, plan and consolidate this study are listed in Table 1.1. The study scope was set by RA-001 through risk ranking, PCP-005 is the approved protocol this report executes and PCMR-001 consolidates the results of all eight per-step reports.

Table 1.1: Documents related to this report.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-005	Process Characterization Plan (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

1.4 Regulatory and scientific basis

The study is a Stage 1 process design activity in the sense of the lifecycle guidance (U.S. Food and Drug Administration 2011). The record of that study is the evidence that the commercial ranges proposed for this step rest on data and on prior knowledge instead of on the transfer of a set-point alone. The design space concept used in §6 is supplied by ICH Q8, the risk methodology carried forward from RA-001 by ICH Q9 and the framing of the control strategy in §10 by ICH Q11 (International Council for Harmonisation 2009, 2023b, 2012).

Criticality is treated as a continuum in this package. Attributes are assigned a level of very low, low, moderate, high or very high, of which the levels high and very high are called critical (CMC Biotech Working Group 2009). Parameter classes follow the same logic: a well-controlled critical process parameter is one whose variability affects a critical quality attribute but which carries a low risk of leaving the design space, whereas a critical process parameter carries a high risk of leaving it. Since

that distinction rests on control capability and on the size of the region as well as on the size of the effect, it is applied at the end of this report (§9) and not at the start.

No viral clearance claim is made for the Protein A step. The modular clearance claim for A-Mab rests on low-pH viral inactivation, anion exchange and small-virus retentive filtration, all consolidated in PCMR-001 under ICH Q5A(R2) (International Council for Harmonisation 2023a).

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The Protein A capture step is a platform operation at Novacyte Biologics with an extensive performance history. It has been used to manufacture three earlier humanized IgG1 products (X-Mab, Y-Mab and Z-Mab) and to supply the A-Mab clinical programme. Across that experience the step has recovered antibody consistently and has reduced host cell protein and DNA by a large factor at every scale operated. This body of experience reduced the amount of development work the present study had to do. Characterization at this step confirms and bounds behaviour that is already understood rather than discovering it, which is the same argument the case study makes for its own downstream platform (CMC Biotech Working Group 2009). The transfer of that credit is justified by mechanism rather than by similarity alone, because the resin chemistry, the bed geometry and the buffer system used for A-Mab are the platform ones, with the antibody itself the only material difference between A-Mab and the earlier products at this step.

The mechanistic expectations the study set out to test were also supplied by prior knowledge. Elution buffer pH was expected to be the dominant parameter for pool host cell protein, because the pH of the elution buffer determines which of the impurities bound to the resin and to the product desorb together with the antibody. Protein load was expected to act in the opposite direction: as the load approaches the dynamic binding capacity of the resin, the mass-transfer zone extends further down the bed and a greater share of the loaded impurity burden is carried into the eluate. The two parameters were expected to interact, since the impurity recruited by a heavy load and the impurity recruited by an aggressive elution are partly the same material.

Step yield was expected to be governed by the end of pool collect, which sets how far down the descending edge of the elution peak the pool is taken. Load flow rate acts on yield through residence time in the bed, since a high linear velocity shortens contact and increases breakthrough during the load phase, and that effect was expected to be strongest when the bed is already heavily loaded.

Leaching of the affinity ligand was expected to be a property of the resin and of the elution condition, and to be small enough over the proposed ranges to be hard to resolve. Leached Protein A is assigned by the platform to the resin life-cycle programme (SOP-2008) as much as to the operating parameters, because ligand release accumulates with cycle number and sanitization history. The analysis reported in §5.3 was framed by that expectation.

2.2 Quality attributes in scope

One critical quality attribute is set by this step: leached Protein A, a process impurity of low to moderate criticality. The acceptance criterion for that attribute is given in Table 2.1 with the Tool #1 score, which is the product of impact and uncertainty in the criticality assessment (CMC Biotech Working Group 2009).

Table 2.1: The critical quality attribute set by Step 5.

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	0-5 ppm	L-M	16

Two further attributes are governed at this step without being formed here, with all three listed together in Table 2.2 in the order they are discussed below.

Table 2.2: Quality attributes governed at Step 5, with acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	0-5 ppm	L-M	16
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

Host cell protein is the most consequential of the three, at moderate to high criticality, because it is the impurity present in the largest amount in the clarified harvest and because it carries the highest Tool #1 score in Table 2.2. The second of the three, leached Protein A, is of low to moderate criticality and is the one attribute this step adds instead of removing. The third, residual DNA, is of very low criticality and is cleared by a wide margin over the train as a whole.

The acceptance criteria in Table 2.2 apply to the drug substance and not to the Protein A pool. An intermediate has to be judged against a criterion that applies to an intermediate. For this reason the two criteria this report uses are defined in §7. In outline, pool host cell protein is judged against an in-process limit carried back from the drug-substance criterion through the clearance the polishing steps deliver. Leached Protein A keeps the drug-substance criterion itself, which is the conservative choice at a capture step.

2.3 Risk-based prioritization and parameter selection

Every Protein A process parameter was assessed in RA-001 by risk ranking and filtering, each of them scored on potential impact to quality attributes and on potential

interaction with other parameters (CMC Biotech Working Group 2009). Parameters were placed into three groups by that assessment: those warranting multivariate evaluation, those whose ranges could be supported by a univariate study and those needing no new study because platform ranges already cover them. That grouping is the scope this report executes. For each of the six parameters the outcome is recorded in Table 4.1.

Of the 6 parameters, 4 were carried into the multivariate designs: protein load, elution buffer pH, load flow rate and end of pool collect. Protein load and elution buffer pH were selected on the strength of the expected effect on pool host cell protein, while load flow rate and end of pool collect were selected on the expected effect on step yield and on the load by flow interaction the platform had already seen. All four were expected to interact, which is the condition under which a univariate study cannot support a range (PDA TR 60 makes the same point about one-factor-at-a-time experimentation).

Operating temperature and bed height were assigned to univariate assessment, since neither acts on the binding chemistry of the step: temperature affects the kinetics of adsorption and the stability of the eluted antibody over a narrow band, and bed height acts through residence time at a fixed linear velocity. Both are held to platform ranges by equipment design and are readily controlled, which is why a one-at-a-time assessment was judged sufficient for them in RA-001.

The ranges themselves were set wider than the routine control ranges so that the study would describe the process over a knowledge space instead of only at the operating point. Each characterization range in Table 4.1 is at least 2.0 times the width of the corresponding normal operating range. The wider ranges also provide the process understanding that is needed to support later movement within the design space.

3 Materials and methods

3.1 Scale-down model and its qualification

A scale-down chromatography system was qualified as a model of the manufacturing-scale step under SOP-1001. The model was designed on the established principles of chromatographic scaling, holding bed height, linear velocity, residence time, load in grams per litre of resin and all buffer volumes normalized to column volumes at their commercial-equivalent values, and column efficiency was verified before each campaign by plate count and peak asymmetry. Runs were executed in the temperature-controlled chromatography chamber EQ-CHR-118, which was within the calibration interval throughout execution and is next due on 2026-05-30.

The scale-down model was compared with manufacturing-scale performance on the attributes that matter for the claims in this report: step yield, pool volume, pool host cell protein and leached Protein A. No difference of practical consequence between the scales was found. The qualification record is filed under SOP-1001, with the model treated as representative for the purpose of defining operating ranges and of supporting the clearance contribution stated in §10. However, a qualification of this kind bounds what the model may be used for. It supports claims about the operating ranges and about the clearance a single cycle delivers, without saying anything about how the resin behaves after many cycles, which is why the resin life-cycle is addressed separately in §11.

Centre-point replicates provide the pure-error term against which lack of fit is tested. The centre points were executed at the set-point condition, 3 of them in the screening design and 4 in the response-surface design. The spread of those replicates is reported in §5.1 and used in the lack-of-fit partition in §5.3.

3.2 Operation

Each run was executed to the operating sequence of SOP-2007: equilibration, load, post-load wash, low-pH elution with pool collection on the descending edge, strip, sanitization and storage. Only the four studied parameters were allowed to vary. Every other setting was held at the commercial-equivalent value, including the wash composition, the equilibration volume and the sanitization sequence.

Materials controlled to platform specification were verified for identity and quality and were not characterized in this study. Resin lots were qualified before use and were tracked under SOP-2008, which also governs packing, cycle counting and sanitization. Buffers were prepared and released against a pH and conductivity specification before

use, a control that is directly relevant to the deviation reported in §13.1. The clarified harvest used as feed was drawn from material characterized in PCR-004.

3.3 Analytical methods

Every response and every released attribute was measured by a validated method. The controlled documents that govern this step are listed in Table 3.1, with the performance figures for the two methods that carry the quantitative claims given in Appendix D.

Table 3.1: Controlled procedures and validated analytical methods for Step 5.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

Pool host cell protein was measured by ELISA under AMV-3012 and is reported as ng of host cell protein per mg of antibody. Leached Protein A was measured by ELISA under AMV-3016 and is reported in ppm, residual DNA by qPCR under AMV-3014 and size variants by SEC-HPLC under AMV-3011. Step yield was calculated from the antibody mass loaded and the antibody mass recovered in the eluate pool.

The two ELISA methods differ in precision by a wide margin, which matters for the interpretation of §5.1 and §5.3. AMV-3012 has an intermediate precision of 9.5 %RSD and AMV-3016 has 6.5 %RSD, against observed effects that differ in relative size by about an order of magnitude between the two attributes. The share of total observed variance attributable to each assay is given in Appendix D and is used in §5.1 to separate analytical scatter from process variation.

3.4 Sampling plan

Samples were taken from the load, the flow-through, the pooled wash and the eluate pool on every run. The eluate pool was assayed for host cell protein, leached Protein A, residual DNA and size variants, while the load and flow-through samples support the mass balance behind the reported step yield.

The centre-point condition was executed 3 times in the screening design and 4 times in the response-surface design, giving 7 independent replicates of the set-point across the campaign. All runs within a design were executed in randomized order under SOP-1002.

3.5 Statistical methods

Factors were coded so that the midpoint of each characterization range is zero and the edges are -1 and $+1$. At this step the midpoint of every characterization range coincides with the set-point of the parameter. Therefore the coded origin and the set-point condition are one and the same.

The screening data were analysed as an ordinary least-squares model in the four main effects and the six two-factor interactions, with an effect reported as twice the coefficient on the coded factor. The response-surface data were analysed as a full quadratic model in the same four factors, which gives 14 terms and leaves 13 residual degrees of freedom. Significance is judged at $\alpha = 0.05$ throughout.

Model adequacy is reported as the coefficient of determination, the adjusted coefficient, the predicted coefficient computed from the PRESS statistic, the model F ratio with the associated p-value, and lack of fit is tested by partitioning the residual sum of squares into a lack-of-fit component and the pure error from the centre-point replicates. A response for which no factor reaches significance is treated as robust over the ranges studied, in which case the factors are kept in the knowledge space as evidence of that robustness.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter drawn inside the normal operating range, and is reported as a one-sided Cpk against the drug-substance criterion. The proven acceptable ranges in §7 are computed from the same fitted response-surface models, in the two forms described there.

4 Study design

4.1 Factors, ranges and the knowledge space

All 6 parameters of the step are given in Table 4.1 with the set-points, normal operating ranges, characterization ranges, study type and final classification. The characterization range is the knowledge-space edge of each parameter and is not itself a proven acceptable range; the proven acceptable ranges are computed per attribute in §7.

Table 4.1: Process parameters of Step 5, with set-point, normal operating range, characterization range, study type and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	30	20-40	10-50	WC- CPP	multivari- ate
Elution buffer pH	pH	3.55	3.4-3.7	3.2-3.9	WC- CPP	multivari- ate
Load flow rate	cm/hr	200	150-250	100-300	KPP	multivari- ate
End of pool collect	CV	2.6	2.3-2.9	2-3.2	KPP	multivari- ate
Operating temperature	°C	22	18-26	15-30	GPP	univariate
Bed height	cm	20	15-25	10-30	GPP	univariate

The letters used for the coded factors in the effect and coefficient tables are given in Table 4.2. All four multivariate parameters appear in both designs, which means that the same letter is used for the same factor throughout this report.

Table 4.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Elution buffer pH	pH	screening + RSM
C	Load flow rate	cm/hr	screening + RSM
D	End of pool collect	CV	screening + RSM

The range of each factor was justified against prior knowledge instead of being set as a fixed multiple of the routine range. Protein load runs from 10 to 50 g/L resin, which

spans the region from a lightly loaded column to a load approaching the dynamic binding capacity of the resin. Elution buffer pH runs from 3.2 to 3.9, covering the band over which the antibody desorbs completely at the low edge and incompletely at the high edge. Load flow rate runs from 100 to 300 cm/hr, which is the velocity range the commercial column can deliver. End of pool collect runs from 2 to 3.2 column volumes, which brackets the descending edge of the elution peak.

4.2 Screening design

The screening was executed as a two-level full factorial in the four multivariate factors with 3 centre points, giving 19 runs in total. A full factorial in four factors estimates every main effect and every two-factor interaction without aliasing, which a fractional design of the same size could not do. The resulting model carries 10 terms against 8 residual degrees of freedom, which is a near-saturated fit. The purpose of the screening stage is to identify which factors matter, with the coded settings and the measured responses of all 19 runs given in Appendix A.

4.3 Response-surface design

The response-surface design was a face-centred central composite design in the same four factors, with 16 factorial runs, 8 axial runs and 4 centre points, giving 28 runs. Because the axial points sit on the faces of the coded cube, every run stayed inside the characterization ranges of Table 4.1 and no run required a setting outside the knowledge space. The added axial points estimate the four quadratic terms, which is what turns a set of resolved effects into a predictive surface. The face-centred form was chosen in preference to a rotatable design for a practical reason: a rotatable central composite in four factors places the axial points beyond the cube, and settings outside the characterization ranges of Table 4.1 would have taken the study outside the knowledge space RA-001 authorized. The response-surface model is the predictive model of this report and the basis of the design space in §6, with the design matrix and the measured responses given in Appendix B.

Both designs were executed on the same scale-down configuration and with the same feed material, which makes the two datasets directly comparable. Neither design was invalidated or re-executed. Because the deviations recorded in §13 were dispositioned as retained, every run executed under this plan is included in the analysis reported below.

4.4 Univariate assessment

Operating temperature and bed height were assessed one parameter at a time over the characterization ranges in Table 4.1, with the other parameters held at the set-point. Neither was included in the multivariate designs, for the reasons given in §2.3. Both

were classified as general process parameters (Table [4.1](#)). Under SOP-4001 a general process parameter is one whose variation over the range studied was not linked to a critical quality attribute. The classification rationale for the two parameters is set out in §9.

5 Results

5.1 Centre-point performance and reproducibility

The centre-point replicates set the precision against which every effect in this section is judged, and the three responses differ sharply in that precision. The mean, standard deviation and coefficient of variation of each response over the 4 centre points of the response-surface design are given in Table 5.1.

Table 5.1: Centre-point reproducibility in the response-surface design, and the pure-error basis for the lack-of-fit tests in §5.3.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	7.56e+03	357	4.72
Step yield	4	0.946	0.0187	1.97
Leached Protein A (ppm)	4	2.86	0.967	33.8

Step yield is the most reproducible response, at 2.0 %, with pool host cell protein next at 4.7 %, which is small against effects that run to several thousand ng/mg. Leached Protein A is the least precise of the three by a wide margin, at a coefficient of variation of 33.8 % over the same 4 replicates (Table 5.1). That coefficient of variation is larger than the validated intermediate precision of AMV-3016 alone would produce (6.5 %RSD, Appendix D). Run-to-run variation in ligand release therefore contributes to the centre-point spread alongside the assay.

The screening design gives a second, independent read of the same condition, which is reported in Table 5.2 over 3 replicates.

Table 5.2: Centre-point reproducibility in the screening design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	3	8.1e+03	1.61e+03	19.9
Step yield	3	0.935	0.00599	0.64
Leached Protein A (ppm)	3	3.28	0.508	15.5

Both designs place step yield as the most reproducible response (Table 5.1, Table 5.2), although they disagree on pool host cell protein, where the screening centre points scatter at 19.9 % against 4.7 % in the response-surface design. With only 3 replicates the screening estimate is itself imprecise, which is why the lack-of-fit tests in §5.3 use the response-surface pure error.

5.2 Screening: factor effects

The factors that govern pool host cell protein and step yield were resolved in the screening stage. By contrast, nothing at all was resolved for leached Protein A. The fit of each screening model is given in Table 5.3.

Table 5.3: Screening model fit for the three responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.973	0.939	0.886	28.6	3.49e-05	1.4e+03
Step yield	19	0.976	0.945	0.798	32	2.28e-05	0.00833
Leached Protein A (ppm)	19	0.371	-0.414	-2.98	0.473	0.867	0.513

The models for pool host cell protein and step yield fit well, whereas the model for leached Protein A does not. Pool host cell protein and step yield reach R² of 0.973 and 0.976 with model p-values below $\alpha = 0.05$, whereas leached Protein A reaches R² = 0.371 with an adjusted R² of -0.414 and a model p-value of 0.87. A negative adjusted coefficient on a near-saturated fit means the terms explain less than their degrees of freedom cost, which is the signature of a null result.

Pool host cell protein was affected by elution buffer pH, by protein load and by the interaction between them. The effect estimates for pool host cell protein are given in Table 5.4, in descending order of magnitude.

Table 5.4: Screening effect estimates for pool host cell protein, the six largest by magnitude.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	
D	707.9	353.9	350	1.01	0.341	

Raising elution buffer pH across the characterized range lowered pool host cell protein by 8,840 ng/mg, while raising protein load raised it by 7,269 ng/mg. Both effects are large against a centre-point standard deviation of a few hundred ng/mg, at $p < 0.05$. The third of the significant effects is the load by pH interaction, at -2,587 ng/mg ($p = 0.006$). The negative sign means that the penalty for loading heavily is smaller when the elution is run at the high end of the pH range. 3 of the 10 terms reach significance, with every term involving load flow rate among the 7 that do not.

Step yield was affected by more factors and by smaller amounts. The dominant factor for step yield is end of pool collect, at 0.060, which is a gain of about 6.0 percentage

points of yield across the collect range. Protein load and load flow rate each reduce yield, by 0.026 and 0.024 respectively, and the interaction between them is significant at -0.021 ($p = 0.0010$). The six largest estimates for step yield are given in Table 5.5.

Table 5.5: Screening effect estimates for step yield, the six largest by magnitude.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*
BD	-0.00981	-0.0049	0.00208	-2.36	0.0463	*

Leached Protein A returned no significant term at all. In Table 5.6 the largest of the six estimates is the load by flow interaction, at 0.329 ppm with a p-value of 0.24.

Table 5.6: Screening effect estimates for leached Protein A, the six largest by magnitude. None reaches significance.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AC	0.329	0.165	0.128	1.28	0.235	
A	-0.208	-0.104	0.128	-0.811	0.441	
BD	-0.189	-0.0944	0.128	-0.736	0.482	
BC	-0.182	-0.0912	0.128	-0.712	0.497	
C	0.18	0.0898	0.128	0.701	0.503	
D	0.152	0.0758	0.128	0.591	0.571	

Every estimate in Table 5.6 is smaller than the centre-point standard deviation of the same response. No factor could therefore be separated from the run-to-run noise on this design. That null result is informative in itself. The four factors stay in the knowledge space as evidence that ligand release is insensitive to them over the ranges screened, which is the basis on which they are classified in §9.

These effect magnitudes identify which factors matter, but they are not the predictive model of this report, because a near-saturated two-level fit with 8 residual degrees of freedom can neither separate curvature from error nor predict at an interior point of the region. Those estimates are refined by the response-surface models in §5.3.

5.3 Response-surface models

The response-surface stage produced one model that is adequate and predictive, one that is adequate, and one that is neither. The fit of all three is given in Table 5.7.

Table 5.7: Response-surface model fit for the three responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.989	0.976	0.932	80.9	2.71e-10	895
Step yield	28	0.913	0.818	0.586	9.69	0.000105	0.016
Leached Protein A (ppm)	28	0.567	0.101	-0.714	1.22	0.365	0.637

Pool host cell protein is modelled well. The full quadratic model reaches $R^2 = 0.989$, adjusted $R^2 = 0.976$ and predicted $R^2 = 0.932$, with a model F of 80.9 ($p = 2.7e-10$). A predicted coefficient close to the adjusted coefficient means the model predicts a held-out run about as well as it fits the data used to build the model, which is the property the design space in §6 depends on. Step yield is adequate but less predictive, at $R^2 = 0.913$ and predicted $R^2 = 0.586$, which is enough to describe the yield surface and to support the classification of the two key process parameters in §9.

Leached Protein A yielded no usable model. The quadratic fit reaches $R^2 = 0.567$ (Table 5.7), but the adjusted coefficient falls to 0.101 and the predicted coefficient is -0.714. A negative predicted coefficient means the model predicts a held-out run worse than the mean of the data predicts. The model F ratio is 1.22 at $p = 0.36$, which is to say that the model as a whole explains no more than chance would. This report therefore makes no predictive use of it, with the treatment of the attribute set out below.

The coefficient table for pool host cell protein is given in Table 5.8, in which 7 of the 14 terms reach significance at $\alpha = 0.05$.

Table 5.8: Full quadratic coefficients for pool host cell protein, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	7,215.5	309	23.3	5.44e-12	***
A	3,871.8	211	18.4	1.12e-10	***
B	-5,244.6	211	-24.9	2.4e-12	***
C	-435.4	211	-2.06	0.0596	
D	-1,055.5	211	-5	0.000242	***
AB	-1,899.8	224	-8.49	1.16e-06	***
AC	-429.1	224	-1.92	0.0774	
AD	-518.1	224	-2.32	0.0376	*
BC	348.2	224	1.56	0.144	
BD	727.4	224	3.25	0.00632	**
CD	-128.7	224	-0.575	0.575	
A ²	-344.5	557	-0.618	0.547	
B ²	1,943.5	557	3.49	0.00401	**
C ²	851.3	557	1.53	0.151	
D ²	330.5	557	0.593	0.563	

Elution buffer pH carries the largest coefficient at -5,245 ng/mg per coded unit, with protein load next at 3,872. The interaction of the two is significant at -1,900, reproducing the sign and roughly the magnitude estimated in the screening stage. The quadratic term in elution pH is significant at 1,944 ($p = 0.0040$), which is the curvature a two-level design cannot see. End of pool collect reaches significance in this model at -1,055 ng/mg per coded unit, whereas the screening estimate for the same factor was not significant and carried the opposite sign (Table 5.4). The two estimates do not conflict, because a non-significant screening estimate makes no claim about the sign. Therefore the response-surface estimate governs, in line with the framing set out in §5.2.

The residual for pool host cell protein is partitioned into lack of fit and pure error in Table 5.9.

Table 5.9: Analysis of variance for pool host cell protein, with the residual partitioned against the centre-point pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.08e+08	14	6.48e+07	80.9	2.71e-10
Residual	1.04e+07	13	8.01e+05	nan	nan
Lack of fit	1e+07	10	1e+06	7.88	0.058
Pure error	3.82e+05	3	1.27e+05	nan	nan

Lack of fit does not reach significance at $\alpha = 0.05$ ($F = 7.88$, $p = 0.058$), which supports the quadratic form for this response. However, that p-value is the closest to α of the three responses, on a test that rests on 4 centre-point replicates. For this reason the result bounds the evidence for the model form without establishing it. The design space in §6 is bounded accordingly. The corresponding tests for step yield and leached Protein A are further from significance, at $p = 0.73$ and $p = 0.95$ (Appendix C).

The diagnostics for pool host cell protein are given in Figure 5.1, where the residuals show no structure against the predicted value, the normal plot is close to a straight line and the actual values fall on the identity line across the whole range of the response.

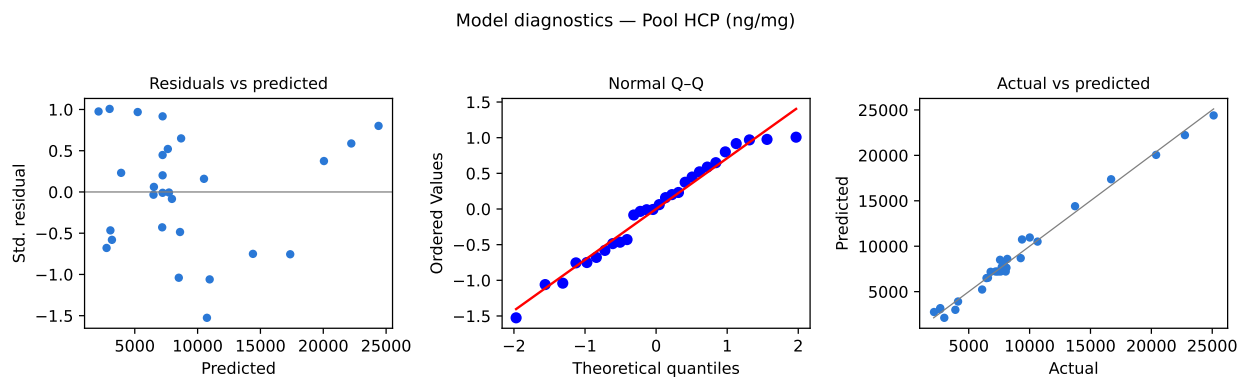


Figure 5.1: Model diagnostics for pool host cell protein: standardized residuals against predicted value, normal quantile plot, and actual against predicted.

The fitted surfaces are shown in Figure 5.2 over protein load and elution buffer pH, with the other two factors held at the design centre. Pool host cell protein rises steeply towards the corner at high load and low pH. The contours are curved instead of parallel, which is the interaction and the pH curvature already seen in Table 5.8. Step yield varies over a narrow band across the same plane, and leached Protein A shows almost no gradient, consistent with a model that explains nothing.

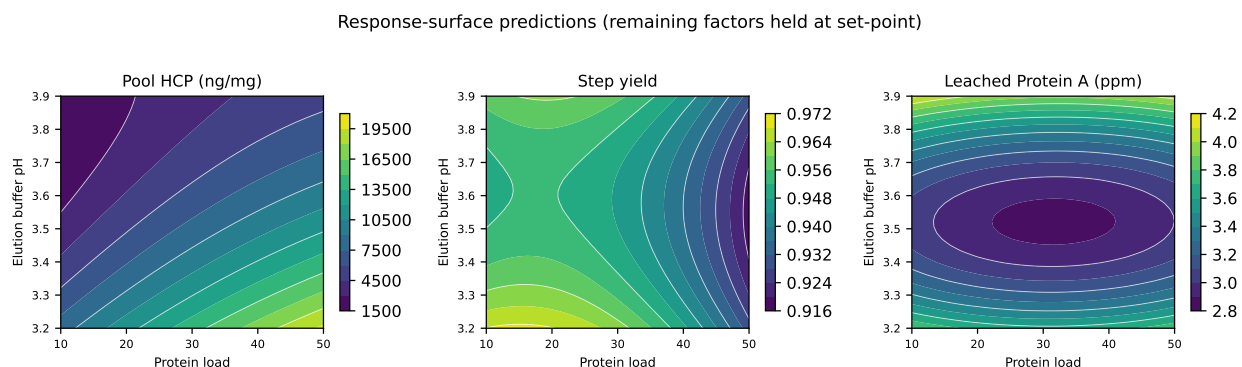


Figure 5.2: Response-surface predictions for the three responses over protein load and elution buffer pH, with load flow rate and end of pool collect held at the design centre.

The leached Protein A model is retained as knowledge-space evidence and is put to no other use in this report. The diagnostics for that model are given in Figure 5.3. Exactly one of the 14 terms reaches significance, the quadratic in elution pH at $p = 0.040$, which is about what testing that many terms at $\alpha = 0.05$ produces by chance. Lack of fit is far from significant ($p = 0.95$, Appendix C). That test compares the residual with the pure error. The pure error for this response is large (33.8 % at the centre point). A non-significant result here therefore says only that the quadratic form is not contradicted by a noisy replicate set, which is not evidence that the model predicts.

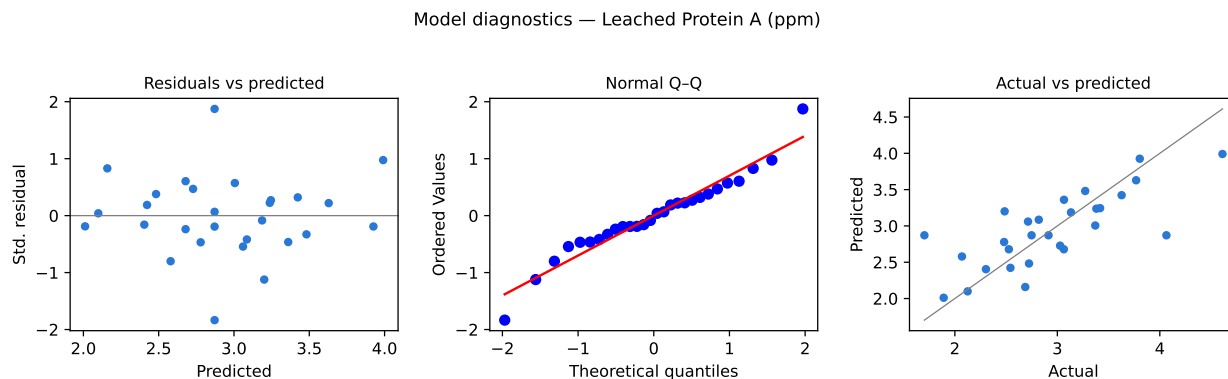


Figure 5.3: Model diagnostics for leached Protein A. The actual against predicted panel shows the scatter that a predicted R-squared below zero describes.

The operative claim for leached Protein A is therefore made from the assayed pools

directly. Across the 47 runs of the two designs, the pools ranged from 1.70 to 4.61 ppm, with a mean of 2.95 ppm and a standard deviation of 0.58 ppm, against an acceptance criterion of 5 ppm, which 47 of them met. The highest single pool left 0.39 ppm of margin, which is narrow on one run and is the reason the attribute is monitored on every batch (§10). However, the criterion applied here is the drug-substance specification, which is applied to the capture pool conservatively. Leached Protein A is cleared downstream of this step by cation exchange and anion exchange (PCR-007, PCR-008). In addition, only 0.003 ppm of leached Protein A is carried by the simulated drug substance at commercial scale (§8).

5.4 Mechanistic interpretation

The surfaces in Figure 5.2 follow from the physical chemistry of affinity capture and confirm the expectations recorded in §2.1. Elution buffer pH governs pool host cell protein because it sets the aggressiveness of desorption: a lower pH releases the antibody faster and also releases the more tightly associated host cell protein, which then co-elutes into the collected pool. The convex quadratic term means the effect is not uniform across the range: host cell protein falls steeply as pH rises from the low edge and then flattens above the set-point, which leaves the pool most sensitive to pH exactly where the pH is lowest.

Protein load acts through the capacity of the bed: at a light load the antibody occupies a small fraction of the available ligand and the impurity that binds non-specifically is washed away efficiently, whereas a load approaching dynamic binding capacity extends the mass-transfer zone and retains a greater share of the loaded impurity into the elution. The negative interaction between load and pH follows directly: at a high elution pH part of the impurity recruited by a heavy load is left behind on the resin in any case, which makes the two effects less than additive over the characterized ranges.

Step yield is governed by where the pool is stopped. Collecting further down the descending edge recovers antibody that would otherwise go to strip, which is why end of pool collect carries the dominant positive term. By contrast, load and flow act in the opposite direction through breakthrough during the load phase. The interaction between them is negative (Table 5.5) because a short residence time matters most when the bed is already close to saturation.

Leaching of the affinity ligand behaves as a resin property over these ranges, since ligand release depends on the chemistry of the immobilization, on the cumulative cycle count and on the sanitization history, none of which is a parameter of a single run. Nothing in the data contradicts that account. The four operating parameters together explain no more of the pool variation than chance would, while the centre-point replicates scatter more widely than any factor moved the response.

These results also explain why a multivariate region is required in place of four independent ranges. The two parameters that bound the operating region interact, with the surface curved in one of them. A univariate range for protein load would have to be set at the load that is acceptable at the worst elution pH. A univariate

range for elution pH would have to be set at the pH that is acceptable at the heaviest load. Consequently, both would be narrower than the process needs, whereas the multivariate region in §6 states the constraint the process actually faces.

6 Design space

The design space for Step 5 is defined as the region in protein load, elution buffer pH, load flow rate and end of pool collect over which the fitted response-surface models predict a pool host cell protein at or below 10,724 ng/mg and a leached Protein A at or below 5 ppm. Evaluated on an even grid of 14,641 settings across the characterized ranges, 73.4% of the region meets both criteria. The binding constraint on that region is Pool HCP (ng/mg), since leached Protein A rejects 0.0% of the grid on its own.

The principal plane of the region is protein load against elution buffer pH (the plane drawn in Figure 5.2). Load flow rate and end of pool collect do not bound the region within the characterized ranges. The role of load flow rate and end of pool collect in the control strategy is to govern step yield (§10). At the set-point the model predicts 7,216 ng/mg, which is 1.49-fold below the in-process limit.

The worst case predicted by the same model within the characterized region is the corner at protein load 50 g/L resin, elution buffer pH 3.2, load flow rate 100 cm/hr and end of pool collect 2 CV, at 24,397 ng/mg. That corner lies far outside the design space and is an edge of failure for the step rather than a setting the process may take. 9 of the 47 executed runs produced a pool above the in-process limit, all 9 of them at the low edge of the elution pH range. Those runs are the direct experimental evidence for where the boundary lies.

The normal operating range is not wholly contained in the design space, and that gap has to be stated plainly. 6.2% of the settings inside the normal operating range exceed the in-process limit on the fitted model, the worst of them being the corner at protein load 40 g/L resin, elution buffer pH 3.4, load flow rate 150 cm/hr and end of pool collect 2.3 CV, at which the model predicts 13,553 ng/mg. Three considerations bound what that means. The in-process limit is itself set 4.5-fold below the pool concentration of 48,260 ng/mg at which the downstream train would no longer bring a batch into specification, which makes an excursion above the limit an alert instead of a batch failure. At that same corner the batch would still reach specification, with the drug substance at 28 ng/mg against a limit of 100 ng/mg, on the clearance the two polishing steps deliver in the nominal train. And the corner requires both parameters to sit at an unfavourable edge at the same time.

The operative constraint is therefore a joint one. Protein load and elution buffer pH must be held so that the pair lies inside the region, with neither read as an independent range at its own edge. At the top of the load normal operating range, 40 g/L resin, elution buffer pH must be at least 3.46 with the other two factors at their set-points. That value sits above the low edge of the pH normal operating range, 3.4, which is the same finding expressed as a pH.

Movement within the design space is not considered a change in the sense of ICH Q8, although all changes remain subject to the pharmaceutical quality system (ICH Q10). Movement outside the design space requires change control with a supporting assessment (International Council for Harmonisation 2009, 2008). Three bounds apply to the claim. The region is defined only over the characterization ranges in Table 4.1 and says nothing about settings beyond them. The models predict mean levels and carry no interval, which is why the assurance form of the claim is the NOR-propagated analysis in §7. And the whole region rests on the scale-down model qualified in §3.1, which was qualified on performance attributes and not on resin life-cycle behaviour. In practice the region is used in two ways: a proposed change to a set-point or a range is checked against the region before execution, and any excursion recorded during manufacture is assessed against it as part of the batch disposition.

7 Proven acceptable ranges

The proven acceptable ranges complement the multivariate region with a per-attribute, per-parameter view. Since the acceptance basis differs between the two governed responses, and since the ranges cannot be read without knowing which criterion produced them, the two criteria are defined before the table. Pool host cell protein is judged against an in-process limit of 10,724 ng/mg, carried back from the drug-substance criterion of 100 ng/mg through the clearance credited to cation exchange (78-fold) and anion exchange (6.2-fold), then divided by an assurance factor of 4.5. Leached Protein A has no in-process limit configured and is judged against the drug substance criterion of 5 ppm directly.

Both analyses are given in Table 7.1 for every governed attribute and every multivariate parameter, and the first column of ranges holds the other three factors at their set-points. The second propagates the normal operating range of the other three by Monte-Carlo of the fitted model and requires the 95 % predictive interval to stay within the criterion, which makes it a robustness statement instead of a mean-level one.

Table 7.1: Proven acceptable ranges per governed attribute and parameter, at set-point and with the other parameters varying within their normal operating ranges.

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Protein load	10-50	g/L resin	$\leq$ 1.07e+04	10-49.8	10-34.5
Pool HCP (ng/mg)	Elution buffer pH	3.2-3.9	pH	$\leq$ 1.07e+04	3.36-3.9	3.48-3.9
Pool HCP (ng/mg)	Load flow rate	100-300	cm/hr	$\leq$ 1.07e+04	100-300	138-300
Pool HCP (ng/mg)	End of pool collect	2-3.2	CV	$\leq$ 1.07e+04	2-3.2	2.29-3.2
Leached Protein A (ppm)	Protein load	10-50	g/L resin	$\leq$ 5	10-50	10-50
Leached Protein A (ppm)	Elution buffer pH	3.2-3.9	pH	$\leq$ 5	3.2-3.9	3.2-3.86
Leached Protein A (ppm)	Load flow rate	100-300	cm/hr	$\leq$ 5	100-300	100-300

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Leached Protein A (ppm)	End of pool collect	2-3.2	CV	≤ 5	2-3.2	2-3.2

Leached Protein A is acceptable across the whole characterization range of every parameter under the at-set-point analysis, and under NOR propagation for three of the four. The one exception is elution buffer pH, whose NOR-propagated range narrows to 3.2–3.86 against a characterization range of 3.2–3.9. The excluded sliver lies at the high pH edge and follows from the quadratic term discussed in §5.3, which the analysis propagates because it is in the fitted model. Given that the model explains no more than chance, this narrowing should be read as an artefact of propagating a null model instead of as a demonstrated pH effect. Over the whole normal operating range the same model predicts a maximum of 3.24 ppm.

Pool host cell protein behaves differently, and the NOR-propagated ranges for that response are the binding result of this section. Under the at-set-point analysis three parameters are acceptable across their whole characterization ranges and elution buffer pH is acceptable from 3.36–3.9. Under NOR propagation every range narrows: protein load to 10–34.5 g/L resin against a normal operating range of 20–40, and elution buffer pH to 3.48–3.9 against a normal operating range of 3.4–3.7. Neither of the two proven acceptable ranges covers the whole of the corresponding routine range.

That result is the univariate expression of the design-space finding in §6 and carries the same consequence. The upper part of the load normal operating range is proven only when elution buffer pH sits above the low part of the pH range. In the same way, the low part of the pH range is proven only at a load below the top of the load range. Load flow rate and end of pool collect behave differently again. Both of the NOR-propagated ranges, 138–300 cm/hr and 2.29–3.2 CV, contain the whole of the corresponding normal operating range. As a result, neither of the two constrains the operating region.

The governing parameter for pool host cell protein is shown in Figure 7.1 and the governing parameter for leached Protein A in Figure 7.2. In each figure the acceptable region is shaded, the acceptance limit is drawn and the set-point and normal operating range are marked.

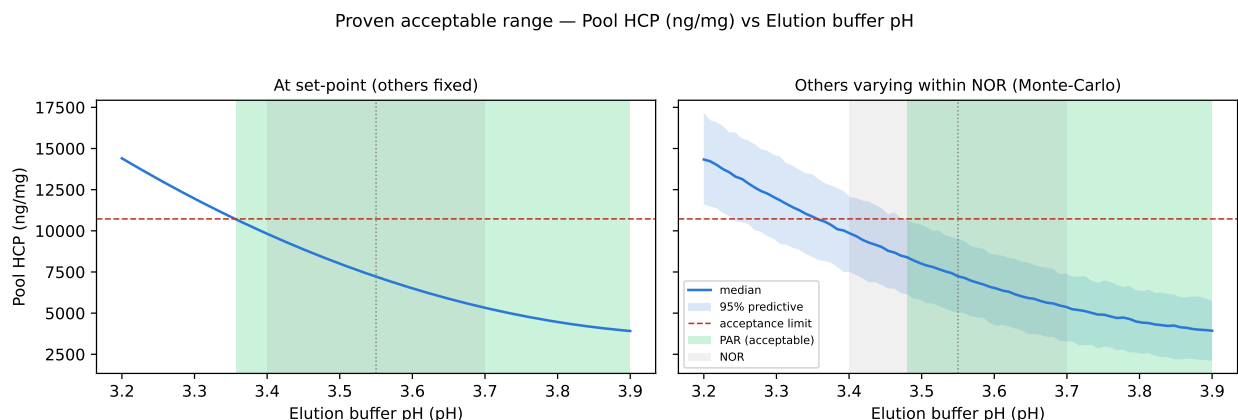


Figure 7.1: Proven acceptable range for pool host cell protein against elution buffer pH, at set-point (left) and with the other parameters varying within their normal operating ranges (right).

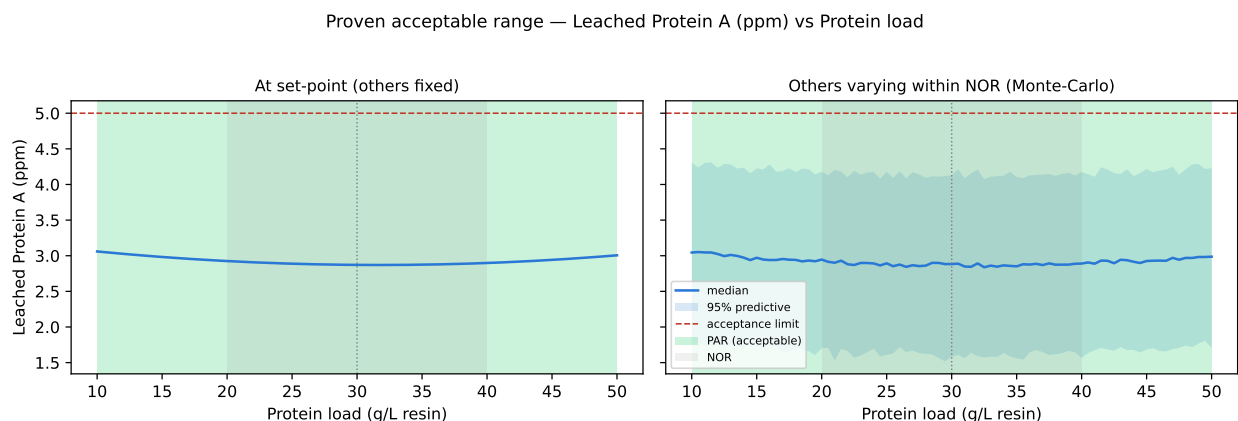


Figure 7.2: Proven acceptable range for leached Protein A against protein load, at set-point (left) and with the other parameters varying within their normal operating ranges (right).

The relationship between these ranges and the design space should be read in one direction only. A proven acceptable range is a univariate or NOR-robust statement about one parameter, whereas the design space of §6 is the multivariate region. Where a whole characterization range is proven acceptable, as it is for leached Protein A against three parameters, the binding constraint on the operating region is the multivariate corner and not any univariate range. Where a proven acceptable range is narrower than the routine range, as it is for protein load and elution buffer pH against pool host cell protein, the narrower result governs. Both statements are bounded by the characterization ranges in Table 4.1 and by the fitted models of §5.3. Both analyses also share the limitation of the model they are computed from, since a proven acceptable range derived from a fitted surface can be no more reliable than that surface, which is why the leached Protein A rows of Table 7.1 are reported without being relied on.

8 Process capability and robustness

Capability at commercial scale was estimated by drawing 2,000 simulated batches with every parameter distributed inside the normal operating range and propagating them through the process model. The resulting distribution of each governed attribute is reported in Table 8.1 against the drug-substance criterion, as a one-sided Cpk. Because the simulation propagates parameter variation through the fitted models of every step in the train, those figures are a statement about the process as a whole and not about this step alone.

Table 8.1: Commercial-scale capability of the attributes governed at Step 5, from Monte-Carlo simulation of NOR-bounded batches.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 $\pm$ 4.6	6.1
Residual DNA	VL	upper	0-0.001	0.0001 $\pm$ 0	10.8
Leached Protein A	L-M	upper	0-5	0.003 $\pm$ 0.0006	2,785.3

Host cell protein is the tightest of the three, at Cpk 6.1. The simulated mean is 15.2 ng/mg with a standard deviation of 4.6. The largest of the 2,000 simulated batches reached 42.3 ng/mg against a specification of 100 ng/mg. That capability is not delivered by this step alone: Protein A capture takes the harvest burden down to about 9,125 ng/mg in the nominal train, after which cation exchange (PCR-007) and anion exchange (PCR-008) contribute 78-fold and 6.2-fold in turn.

The other two attributes sit far from the limits. Residual DNA reaches Cpk 10.8 (Table 8.1), with this step contributing 2.5 log₁₀ of clearance and anion exchange contributing the rest (PCR-008). Leached Protein A reaches Cpk 2,785 at a simulated mean of 0.003 ppm, which is 1,667-fold below the criterion. An index of that size should be read as showing only that the attribute does not constrain the process, with Table 8.1 reporting it for completeness.

Two bounds apply to these figures. They describe operation within the normal operating range and say nothing about the excursion corners discussed in §6. They also rest on the scale-down model and on the process model behind the simulation, neither of which represents the resin life-cycle. The tightest capability anywhere in the drug substance is 1.51, which belongs to a viral clearance attribute that this step makes no claim on (PCMR-001).

9 Parameter classification

Each parameter was classified under SOP-4001 from the demonstrated effects in §5 and from the risk of leaving the design space in routine operation. The outcome is carried in Table 4.1, with the rationale for each parameter set out below.

- Protein load is classified as a well-controlled critical process parameter. Load carries the second largest coefficient on pool host cell protein and interacts with elution buffer pH (Table 5.8). Because the NOR-propagated proven acceptable range for load, 10–34.5 g/L resin, does not cover the upper part of the normal operating range, the parameter is linked to a critical quality attribute and bounds the operating region. Load is set by a mass calculation from a measured feed titre and is readily controlled, which keeps the risk of leaving the design space low and makes the class well-controlled critical instead of critical.
- Elution buffer pH is classified as a well-controlled critical process parameter. Elution pH carries the largest coefficient of any term on pool host cell protein and the only significant curvature. However, the NOR-propagated proven acceptable range for elution pH, 3.48–3.9, does not cover the low part of the normal operating range. Buffer pH is prepared and released against a specification before use. The deviation in §13.1 shows both that the control can fail and that release testing detects it, which is what a well-controlled parameter means here.
- Load flow rate is classified as a key process parameter, since flow affects step yield significantly and reduces it at high load through the load by flow interaction, while reaching no significant effect on pool host cell protein in either design. The NOR-propagated proven acceptable range for flow, 138–300 cm/hr, covers the whole of the normal operating range. The parameter therefore governs process performance without constraining a quality attribute, which is the key process parameter class.
- End of pool collect is classified as a key process parameter, since collect volume carries the dominant effect on step yield and also reaches significance on pool host cell protein in the response-surface model, at -1,055 ng/mg per coded unit, smaller than the elution pH coefficient by a factor of
 5. Because the NOR-propagated proven acceptable range, 2.29–3.2 CV, covers the whole normal operating range, the effect is real without bounding the region. The parameter is classified as key on that basis and is monitored as a performance parameter.
- Operating temperature is classified as a general process parameter. It was assessed one parameter at a time over 15–30 °C and was not linked to any governed

attribute. In operation it is held within a narrow band by the chromatography chamber EQ-CHR-118.

- Bed height is classified as a general process parameter. It was assessed one parameter at a time over 10–30 cm, acts only through residence time at a fixed linear velocity, and is fixed in operation by the packing acceptance criteria in SOP-2008.

The classification returns 0 critical process parameters at this step, against 2 well-controlled critical, 2 key and 2 general. The absence of a critical process parameter follows from control capability and not from an absence of effect. Both well-controlled critical parameters have a demonstrated effect on a critical quality attribute, while both are held by instrumentation and by released buffers that make an excursion outside the design space unlikely and detectable. In addition, the two key process parameters are held to the normal operating ranges for consistency of yield and pool volume, which is the reason a parameter with no demonstrated link to a quality attribute is nonetheless part of the control strategy (CMC Biotech Working Group 2009).

10 Contribution to the control strategy

Step 5 contributes six elements to the control strategy for the drug substance, stated here in the order a batch encounters them.

Protein load and elution buffer pH are controlled within the normal operating ranges and jointly within the design space of §6. The joint constraint is the operative one, which is why the batch record carries it as a combined limit instead of as two independent ranges. Load flow rate and end of pool collect are controlled within their normal operating ranges as key process parameters, which assures consistent yield and pool volume without being linked to a quality attribute.

Pool host cell protein is monitored on every batch against the in-process limit of 10,724 ng/mg by AMV-3012. A result above that limit is an alert that triggers assessment against the downstream clearance actually available, since the limit sits 4.5-fold below the concentration at which the train can no longer bring a batch into specification.

Leached Protein A is measured on the capture pool of every batch by AMV-3016 and trended against resin cycle number. This is the control that carries the attribute, because the characterization studies found no operating parameter that governs it (§5.3). Resin life-cycle limits, packing acceptance criteria and the sanitization sequence are held under SOP-2008. The batch trend is the evidence that leaching stays inside the criterion as the resin ages.

Elution buffer preparation is controlled by release testing against a pH specification before the buffer is used. The deviation in §13.1 arose in buffer preparation and was detected by that release test, which is the reason the control is named here as a contribution of the step. Finally, the four multivariate parameters are recorded on the batch record with the design-space constraint attached to the pair of quality-linked parameters, which allows a reviewer at disposition to see whether the batch was operated inside the region defined in §6.

The step does not control any of the three governed attributes alone: cation exchange (PCR-007) and anion exchange (PCR-008) deliver the majority of the host cell protein clearance and both clear leached Protein A, while anion exchange delivers the majority of the DNA clearance. No viral clearance is claimed here. The consolidated control strategy, including the cumulative clearance and the parameter classes across the whole train, is given in PCMR-001.

11 Discussion

The step is well understood over the ranges studied, although the understanding is distributed unevenly across the three responses. Pool host cell protein is described by a predictive model with a clear mechanism, an interaction and a curvature, which makes it the response that defines the operating region. Step yield is described adequately and matters for process consistency rather than for quality, whereas leached Protein A is described by no model at all. What is known about the attribute comes from the assayed pools and from the resin life-cycle programme.

The finding that deserves the most attention is the gap between the normal operating range and the design space. 6.2% of the normal operating range exceeds the in-process limit for pool host cell protein on the fitted model, at the corner where a heavy load meets a low elution pH. Three things bound that finding, and none of them removes it: the in-process limit carries an assurance factor of 4.5. The drug substance stays within specification even at that corner, at 28 ng/mg against 100. And the corner requires two parameters to sit at unfavourable edges simultaneously, which routine operation at the set-point does not approach. The response is to state the constraint jointly in §6 and §10, and to keep the in-process monitor as the point of detection.

Confidence in the scale-down model rests on the qualification described in §3.1 and on the agreement between the two independently executed designs. The screening and response-surface models agree on the sign and approximate size of every large effect, having been executed as separate campaigns on the same configuration. The one disagreement, on end of pool collect against pool host cell protein, is between a non-significant screening estimate and a significant response-surface estimate. The disagreement is resolved in favour of the design that can resolve it.

Both deviations recorded in §13 were analytical or preparative in origin and neither altered a dataset. The elution buffer prepared below target delivered a pH of 3.38, which sits inside the characterization range at a coded position of -0.49. The leached Protein A result that was initially out of trend was assayed again at 1.6 ppm. Both runs were retained, with the impact assessments given in §13.

Four limitations apply to the conclusions of this report. The response-surface models predict mean levels and carry no interval of their own, which is why the assurance form of the range claim is the NOR-propagated analysis in §7 and why the design-space boundary should not be treated as a point of high reliability. A single feed state drawn from material characterized in PCR-004 was used in both designs, which leaves the effect of feed variability on this step bounded by that document instead of by this one. Both designs were run at a single resin lot and cycle count, which leaves leaching over the resin lifetime to SOP-2008 and to the batch trend described in §10. And the design space was verified at scale-down and not at commercial scale at the edges of

the region, which is a general limitation of the approach (CMC Biotech Working Group 2009).

No parameter at this step required the most stringent control class. Both quality-linked parameters are held by instrumentation and by released buffers with a low risk of leaving the design space, which is what distinguishes a well-controlled critical process parameter from a critical one under SOP-4001. Overall, the picture is of a step whose quality-relevant behaviour is concentrated in two parameters and whose remaining variation is either analytical or attributable to the resin, which is a simpler control problem than the parameter count of Table 4.1 suggests.

12 Conclusions

Protein A chromatography has been characterized at commercial-equivalent scale-down over 6 parameters, 4 of them in a screening design of 19 runs followed by a response-surface design of 28 runs. The step is well understood and robust within the ranges studied.

A multivariate design space has been defined in protein load, elution buffer pH, load flow rate and end of pool collect, covering 73.4% of the characterized region and bounded by pool host cell protein at high load and low elution pH. Protein load and elution buffer pH must be held jointly inside that region, since neither NOR-propagated proven acceptable range covers the whole of the corresponding routine range.

All 47 capture pools met the leached Protein A criterion of 5 ppm, at a maximum of 4.61 ppm, with no operating parameter identified as governing the attribute. At commercial scale the three governed attributes meet the drug-substance criteria, and the tightest of them is Host Cell Protein (HCP) at Cpk 6.1.

Of the 6 parameters, 2 were classified as well-controlled critical, 2 as key and 2 as general, with 0 classified as critical. Two deviations were investigated and resolved without effect on the models, the region or the classifications.

The step does not deliver the drug-substance host cell protein specification on its own, nor does it carry a viral clearance claim. These conclusions are rolled up into PCMR-001, which consolidates the design spaces, capabilities and classifications of the whole train.

13 Deviations from the plan

Two deviations were recorded during execution of PCP-005, each of them investigated to root cause, impact-assessed against the affected datasets and dispositioned as retained, and Table 13.1 summarizes the two.

Table 13.1: Deviations recorded during execution of PCP-005.

Deviation	Summary	Detected during	Disposition
DEV-005-01	Elution-buffer pH prepared below target on an RSM run	buffer release testing	retained
DEV-005-02	Leached Protein A result initially out-of-trend; re-assayed	analytical review	retained

13.1 Elution buffer prepared below target (DEV-005-01)

An elution buffer lot, LOT-BUF-5290, was prepared below the target pH and was used on one run of the response-surface design. A pH of 3.38 was measured at release testing, against a target of 3.55 for the centre-point condition. Because elution buffer pH carries the largest coefficient of any term on pool host cell protein (\$5.3), a departure in this parameter is the most consequential preparation error the step can make.

The root cause was identified during the investigation as a titration endpoint error in preparation. The pH meter was within calibration, the buffer components were within specification and the endpoint was called early by the operator. Retraining and a second-person verification at the endpoint were implemented for all subsequent buffer preparations.

The impact assessment turned on where the delivered pH falls. A pH of 3.38 sits inside the characterization range of 3.2–3.9 at a coded position of -0.49. Consequently, the run remained inside the region the response-surface model covers, with no extrapolation introduced. The affected run was analysed at the delivered setting and the deviation was dispositioned as retained. Lack of fit for pool host cell protein remains non-significant (Table 5.9), which is the statistical evidence that the affected run is consistent with the fitted surface.

13.2 Leached Protein A result initially out of trend (DEV-005-02)

A leached Protein A result was reported out of trend against the other pools of the campaign and was raised at analytical review. The value fell outside the range the remaining pools occupied, which for this response is a wide range: the campaign spans 1.70 to 4.61 ppm with a centre-point coefficient of variation of 33.8 %.

The probable cause was identified in the assay sequence as suspected carry-over between adjacent wells. The sample was assayed again under AMV-3016 and returned 1.6 ppm, which meets the acceptance criterion of 5 ppm by a wide margin. The validated intermediate precision of the method is 6.5 %RSD with a limit of quantitation of 0.25 ppm, and 41 % of the total observed variance in this response is attributable to the assay itself (Appendix D, Table [14.10](#)). Sequence controls and a bracketing blank were added to the routine sequence of the method.

The run was retained, and the impact on this report is limited by the treatment of the attribute in §5.3, since no predictive model is fitted to leached Protein A and the operative claim is that all 47 assayed pools met the criterion of 5 ppm. A re-assayed result of 1.6 ppm does not approach that criterion and does not change the conclusion. Neither of the two deviations altered a classification, a proven acceptable range or the design space.

14 Appendices A-D

14.1 Appendix A — Screening design matrix and responses

The coded settings and measured responses of all 19 screening runs are given in Table 14.1. Factor letters follow Table 4.2 and run types are factorial or centre.

Table 14.1: Screening design matrix in coded factors, with measured responses.

Run	Type	A	B	C	D	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
1	facto- rial	-1	-1	-1	-1	9,965.7	0.918	2.64
2	facto- rial	-1	-1	-1	1	9,515.4	0.99	3
3	facto- rial	-1	-1	1	-1	9,022.6	0.928	3.03
4	facto- rial	-1	-1	1	1	7,893.7	0.98	2.98
5	facto- rial	-1	1	-1	-1	3,211.7	0.915	2.88
6	facto- rial	-1	1	-1	1	2,523.1	0.986	3.82
7	facto- rial	-1	1	1	-1	2,540.2	0.929	3.3
8	facto- rial	-1	1	1	1	3,108.1	0.96	2.43
9	facto- rial	1	-1	-1	-1	16,568.2	0.913	2.47
10	facto- rial	1	-1	-1	1	19,737.9	0.99	2.66
11	facto- rial	1	-1	1	-1	18,271.3	0.857	2.67
12	facto- rial	1	-1	1	1	21,244.6	0.936	3.54
13	facto- rial	1	1	-1	-1	6,791.9	0.911	2.77
14	facto- rial	1	1	-1	1	7,875.0	0.976	2.29
15	facto- rial	1	1	1	-1	7,655.0	0.891	2.88

Run	Type	A	B	C	D	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
16	facto- rial	1	1	1	1	7,791.9	0.927	3.14
17	center	0	0	0	0	6,436.2	0.931	2.91
18	center	0	0	0	0	9,652.2	0.933	3.07
19	center	0	0	0	0	8,221.9	0.942	3.86

14.2 Appendix B — Response-surface design matrix and responses

The coded settings and measured responses of all 28 response-surface runs are given in Table 14.2. Axial runs sit on the faces of the coded cube.

Table 14.2: Response-surface design matrix in coded factors, with measured responses.

Run	Type	A	B	C	D	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
1	facto- rial	-1	-1	-1	-1	10,011.3	0.93	3.42
2	facto- rial	-1	-1	-1	1	9,269.0	0.99	3.06
3	facto- rial	-1	-1	1	-1	10,650.9	0.928	2.53
4	facto- rial	-1	-1	1	1	7,715.2	0.988	2.48
5	facto- rial	-1	1	-1	-1	2,992.5	0.923	3.27
6	facto- rial	-1	1	-1	1	2,149.1	0.977	1.89
7	facto- rial	-1	1	1	-1	2,643.4	0.908	4.61
8	facto- rial	-1	1	1	1	2,663.4	0.99	3.13
9	facto- rial	1	-1	-1	-1	25,114.3	0.89	3.38
10	facto- rial	1	-1	-1	1	20,388.5	0.99	2.49
11	facto- rial	1	-1	1	-1	22,755.7	0.861	2.12
12	facto- rial	1	-1	1	1	16,694.1	0.924	3.03

Run	Type	A	B	C	D	Pool HCP (ng/mg)	Step yield	Leached Protein A (ppm)
13	facto- rial	1	1	-1	-1	7,882.2	0.895	3.63
14	facto- rial	1	1	-1	1	6,577.7	0.97	2.72
15	facto- rial	1	1	1	-1	6,798.5	0.844	3.06
16	facto- rial	1	1	1	1	6,100.0	0.962	2.82
17	axial	-1	0	0	0	3,900.5	0.935	2.71
18	axial	1	0	0	0	9,377.3	0.935	3.37
19	axial	0	-1	0	0	13,731.8	0.966	3.77
20	axial	0	1	0	0	4,121.9	0.955	3.8
21	axial	0	0	-1	0	7,571.5	0.955	2.3
22	axial	0	0	1	0	8,097.7	0.955	2.54
23	axial	0	0	0	-1	8,167.0	0.908	2.07
24	axial	0	0	0	1	6,460.6	0.953	2.69
25	center	0	0	0	0	8,036.2	0.94	2.75
26	center	0	0	0	0	7,616.5	0.967	1.7
27	center	0	0	0	0	7,206.4	0.924	4.06
28	center	0	0	0	0	7,396.1	0.954	2.91

14.3 Appendix C — Full effect and coefficient tables

Table 14.3 to Table 14.5 give every screening effect estimate for the three responses. Table 14.6 and Table 14.7 give the full quadratic coefficients for the two responses not tabulated in §5.3, and Table 14.8 and Table 14.9 give the corresponding analysis of variance.

Table 14.3: All screening effect estimates for pool host cell protein.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	
D	707.9	353.9	350	1.01	0.341	
BD	-433.1	-216.5	350	-0.619	0.553	
C	167.3	83.7	350	0.239	0.817	
CD	-70.6	-35.3	350	-0.101	0.922	
BC	6.1	3.0	350	0.00868	0.993	

Table 14.4: All screening effect estimates for step yield.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*
BD	-0.00981	-0.0049	0.00208	-2.36	0.0463	*
AB	0.00443	0.00221	0.00208	1.06	0.319	
AD	0.0037	0.00185	0.00208	0.889	0.4	
BC	0.0035	0.00175	0.00208	0.84	0.425	
B	-0.00236	-0.00118	0.00208	-0.568	0.586	

Table 14.5: All screening effect estimates for leached Protein A.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
AC	0.329	0.165	0.128	1.28	0.235	
A	-0.208	-0.104	0.128	-0.811	0.441	
BD	-0.189	-0.0944	0.128	-0.736	0.482	
BC	-0.182	-0.0912	0.128	-0.712	0.497	
C	0.18	0.0898	0.128	0.701	0.503	
D	0.152	0.0758	0.128	0.591	0.571	
AB	-0.133	-0.0664	0.128	-0.518	0.618	
CD	-0.0987	-0.0494	0.128	-0.385	0.71	
B	0.0652	0.0326	0.128	0.254	0.806	
AD	0.0567	0.0284	0.128	0.221	0.83	

Table 14.6: Full quadratic coefficients for step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.947	0.00554	171	3.5e-23	***
A	-0.0165	0.00377	-4.36	0.000767	***
B	-0.00243	0.00377	-0.643	0.531	
C	-0.00891	0.00377	-2.36	0.0346	*
D	0.0365	0.00377	9.68	2.62e-07	***
AB	0.00266	0.004	0.665	0.517	
AC	-0.00914	0.004	-2.28	0.0399	*
AD	0.00629	0.004	1.57	0.14	
BC	0.00243	0.004	0.606	0.555	
BD	0.00291	0.004	0.727	0.48	
CD	0.00219	0.004	0.547	0.594	
A ²	-0.0128	0.00997	-1.29	0.221	
B ²	0.012	0.00997	1.2	0.252	

Term	Coef.	Std. err.	t	p-value	Sig.
C ²	0.00681	0.00997	0.683	0.507	
D ²	-0.0178	0.00997	-1.78	0.0976	

Table 14.7: Full quadratic coefficients for leached Protein A.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	2.87	0.22	13	7.67e-09	***
A	-0.027	0.15	-0.18	0.86	
B	0.148	0.15	0.989	0.341	
C	0.00918	0.15	0.0612	0.952	
D	-0.21	0.15	-1.4	0.185	
AB	-0.0127	0.159	-0.0796	0.938	
AC	-0.143	0.159	-0.899	0.385	
AD	0.132	0.159	0.831	0.421	
BC	0.269	0.159	1.69	0.115	
BD	-0.226	0.159	-1.42	0.179	
CD	0.167	0.159	1.05	0.314	
A ²	0.162	0.397	0.408	0.69	
B ²	0.906	0.397	2.29	0.0397	*
C ²	-0.458	0.397	-1.15	0.269	
D ²	-0.502	0.397	-1.27	0.228	

Table 14.8: Analysis of variance for step yield.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.0348	14	0.00249	9.69	0.000105
Residual	0.00333	13	0.000257	nan	nan
Lack of fit	0.00229	10	0.000229	0.659	0.731
Pure error	0.00104	3	0.000348	nan	nan

Table 14.9: Analysis of variance for leached Protein A.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	6.9	14	0.493	1.22	0.365
Residual	5.27	13	0.406	nan	nan
Lack of fit	2.47	10	0.247	0.264	0.953
Pure error	2.8	3	0.935	nan	nan

14.4 Appendix D — Analytical methods summary

Table 14.10 gives the validated performance of the two methods that carry the quantitative claims of this report. The share of variance attributable to the assay is the figure used in §5.1 and §13.2 to separate analytical scatter from process variation.

Table 14.10: Validated performance of the analytical methods used for the reported responses.

Method	Analytical procedure	Precision (%RSD)	LoQ	LoQ unit	Accuracy (%)	Assay share of variance
AMV-3016	Leached Protein A by ELISA	6.5	0.25	ppm	97.4	0.41
AMV-3012	Host-Cell Protein by ELISA	9.5	1	ng/mg	95	0.4

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